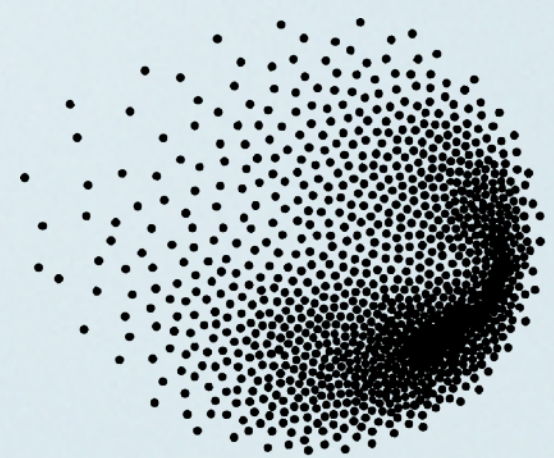
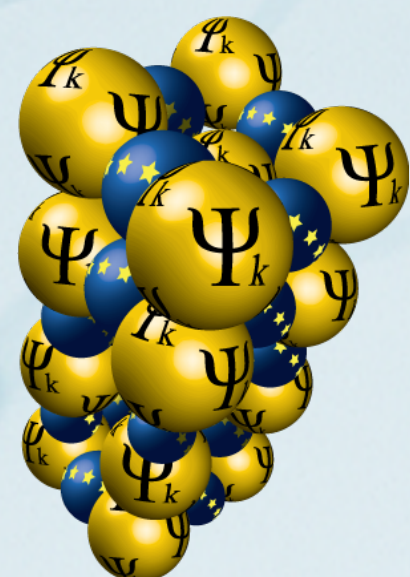


Hands-on: Hybrids, van der Waals, meta-GGA

Presenters: Valentina Sanella (PSI) and Iurii Timrov (PSI)



PSI



ETH4D



Exercise 1

Hybrid functional PBE0 for Si

The $\mathbf{q} + \mathbf{G} = 0$ divergence

In periodic systems, the exact exchange contains a divergence when $\mathbf{q} + \mathbf{G} = 0$:

$$E_x = -\frac{4\pi}{2\Omega} \times \frac{\Omega}{(2\pi)^3} \int d\mathbf{q} \sum_{\mathbf{G}} \frac{A(\mathbf{q} + \mathbf{G})}{|\mathbf{q} + \mathbf{G}|^2}$$

where

We need to setup the $\mathbf{q}$ mesh

We need to setup the $\mathbf{k}$ mesh

$$A(\mathbf{q} + \mathbf{G}) = \frac{\Omega}{(2\pi)^3} \int d\mathbf{k} |\rho_{\mathbf{k},v}^{\mathbf{k}-\mathbf{q},v'}(\mathbf{q} + \mathbf{G})|^2 \equiv \frac{1}{N_k} \sum_{\mathbf{k}} |\rho_{\mathbf{k},v}^{\mathbf{k}-\mathbf{q},v'}(\mathbf{q} + \mathbf{G})|^2$$

(the finite sum over N_k $\mathbf{k}$ -points is what we actually compute) and

$$\rho_{\mathbf{k},v}^{\mathbf{k}-\mathbf{q},v'}(\mathbf{r}) = \psi_{\mathbf{k}-\mathbf{q},v'}^*(\mathbf{r}) \psi_{\mathbf{k},v}(\mathbf{r}).$$

This divergence is integrable, see: F.Gygi and A.Baldereschi, PRB 34, 4405 (1986)

Input file Si.scf.in

```
&control
  calculation='scf'
  restart_mode='from_scratch',
  prefix='Si'
  pseudo_dir = '../..//pseudo'
  outdir='./tmp'
  verbosity='high'
/
&system
 ibrav = 2,
  celldm(1) = 10.20,
  nat = 2,
  ntyp = 1,
  ecutwfc = 20.0,
  input_dft = 'pbe0',
  nqx1 = 1, nqx2 = 1, nqx3 = 1,
  x_gamma_extrapolation = .true.
  exxdiv_treatment = 'gygi-baldereschi'
/
&electrons
  conv_thr = 1.d-9
  mixing_beta = 0.3
/
ATOMIC_SPECIES
Si 28.086 Si.pbe-rrkj.UPF
ATOMIC_POSITIONS {alat}
Si 0.00 0.00 0.00
Si 0.25 0.25 0.25
K_POINTS {automatic}
8 8 8 1 1 1
```

Type of the hybrid functional (PBE0 in this case)

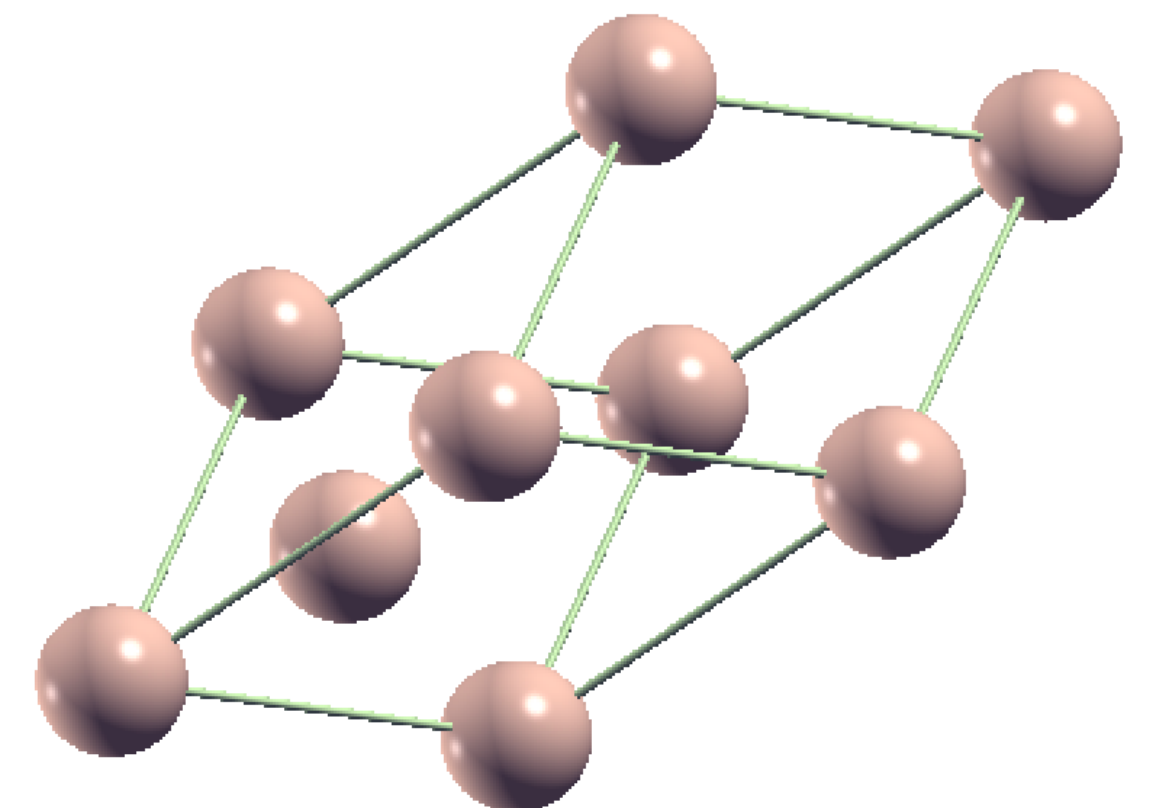
Override the functional written in the pseudopotential (PP) file.
Important: use a PP for the closest GGA (PBE in this case), there are no PP's for hybrids.

q-point mesh

1x1x1 is the minimal mesh ($\mathbf{q}=0$), it is fast but not accurate

Treatment of the $\mathbf{q} + \mathbf{G} = 0$ divergence

Singularity is analytically integrated



Popular hybrid functionals

input_dft = "PBE0"

J.P. Perdew, M. Ernzerhof, K. Burke, JCP 105, 9982 (1996)

C. Adamo, V. Barone, JCP 110, 6158 (1999)

input_dft = "B3LYP"

P.J. Stephens, F.J. Devlin, C.F. Chabalowski, M.J. Frisch, J. Phys. Chem 98, 11623 (1994)

input_dft = "HSE"

Heyd, Scuseria, Ernzerhof, J. Chem. Phys. 118, 8207 (2003)

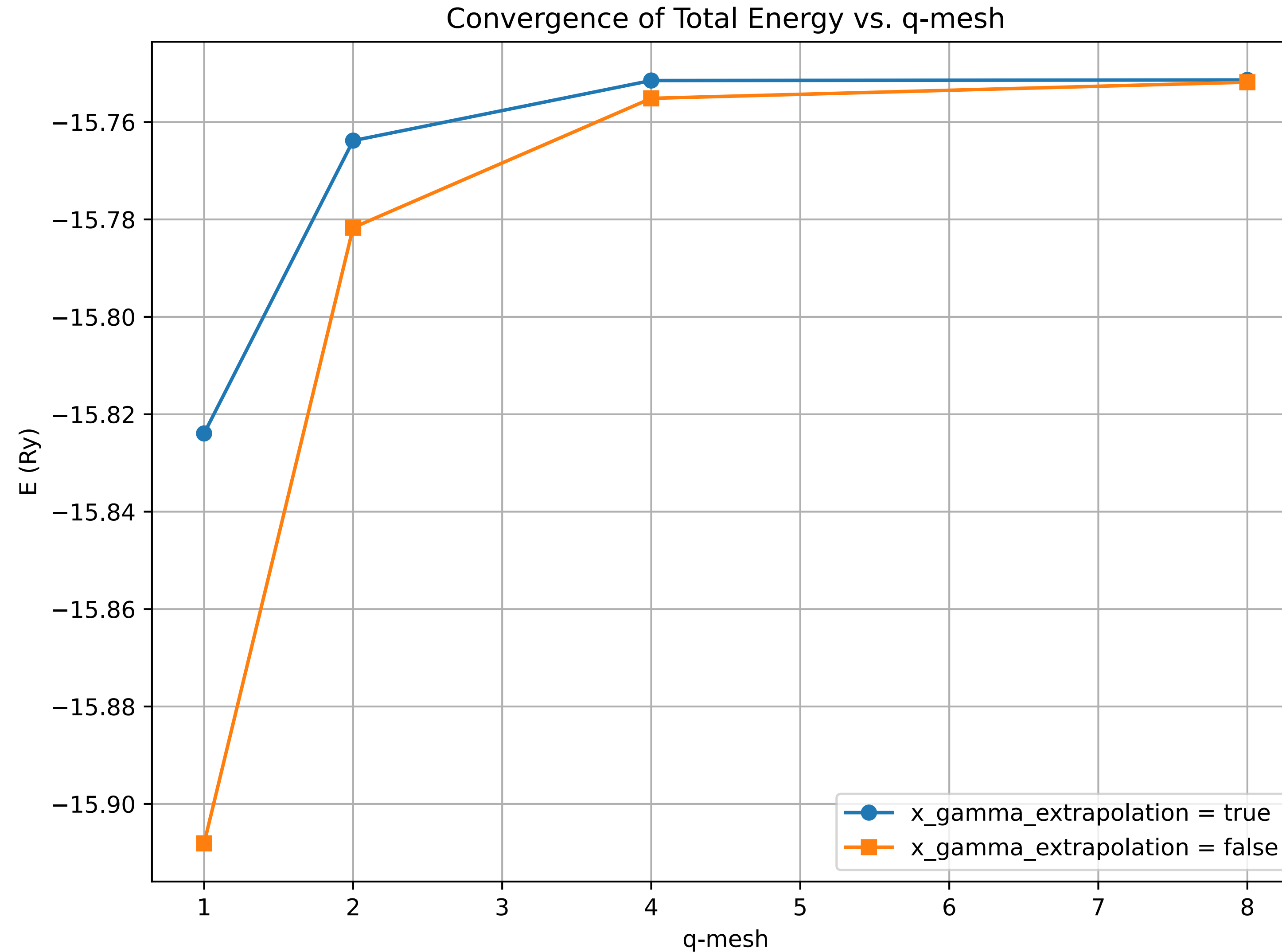
Heyd, Scuseria, Ernzerhof, J. Chem. Phys. 124, 219906 (2006)

Types of treatment of the $\mathbf{q} + \mathbf{G} = 0$ divergence

exxdiv_treatment	CHARACTER
<i>Default:</i>	'gygi-baldereschi'
Specific for EXX. It selects the kind of approach to be used for treating the Coulomb potential divergencies at small $\mathbf{q}$ vectors. 'gygi-baldereschi' : appropriate for cubic and quasi-cubic supercells 'vcut_spherical' : appropriate for cubic and quasi-cubic supercells 'vcut_ws' : appropriate for strongly anisotropic supercells, see also ecutvcut. 'none' : sets Coulomb potential at $\mathbf{G}, \mathbf{q}=0$ to 0.0 (required for GAU-PBE)	

Check the online documentation at https://www.quantum-espresso.org/Doc/INPUT_PW.html

Convergence of the total energy with respect to the q -point mesh

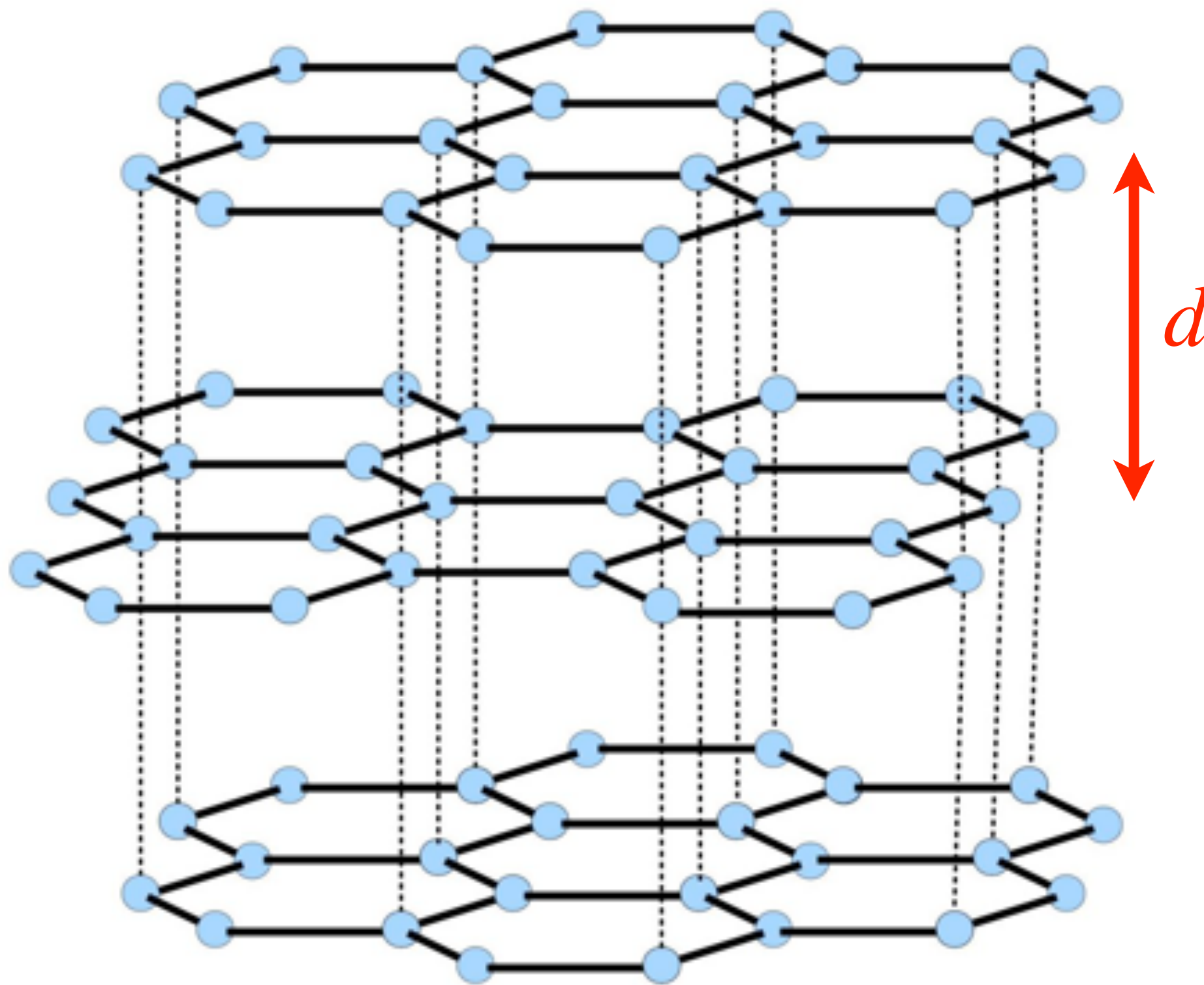


Convergence is faster when $x_gamma_extrapolation = .true$.

Exercise 2

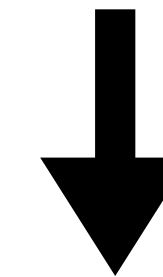
Van der Waals functionals for graphite

Graphite



There are van der Waals (dispersive) interactions between layers in graphite.

The equilibrium inter-layer distance d is too small within LDA and too large within GGA with respect to the experimental value (3.336 Å).



Van der Waals interactions must be taken into account!

Input file graphite.vc-relax.in

```
&control
```

```
  calculation='vc-relax'
```

```
  restart_mode='from_scratch',
```

```
  prefix='graphite'
```

```
  pseudo_dir = '../..pseudo'
```

```
  outdir='./tmp/'
```

```
  verbosity='high'
```

```
  etot_conv_thr = 1.0D-5
```

```
  forc_conv_thr = 1.0D-4
```

```
/
```

```
&system
```

```
 ibrav = 4,
```

```
  a = 2.466,
```

```
  c = 6.411,
```

```
  nat = 4,
```

```
  ntyp = 1,
```

```
  ecutwfc = 30.0,
```

```
  ecutrho = 240.0,
```

```
  occupations = 'smearing',
```

```
  smearing = 'mv',
```

```
  degauss = 0.02,
```

```
  input_dft = 'vdw-DF'
```

```
/
```

```
&electrons
```

```
  conv_thr = 1.d-9
```

```
  mixing_beta = 0.3
```

```
/
```

```
&ions
```

```
/
```

```
&cell
```

```
/
```

```
ATOMIC_SPECIES
```

```
 C 12.011 C.pbe-rrkjus.UPF
```

```
ATOMIC_POSITIONS {crystal}
```

```
 C 0.000000 1.000000 0.75000
```

```
 C 0.666667 0.333333 0.75000
```

```
 C 0.000000 1.000000 0.25000
```

```
 C 0.333333 0.666667 0.25000
```

```
K_POINTS {automatic}
```

```
 4 4 2 1 1 1
```

Variable-cell optimization

Hexagonal lattice, a and c are in Angstrom!

Type of the van Der Waals functional (vdw-DF in this case)

Override the functional written in the PP file.

Important: use PP for the closest GGA (PBE in this case),
because there are no PP for non-local functionals

The **k** point mesh is denser along X and Y than along Z (in
reciprocal space), reflecting shorter periodicity in the XY plane

Types of the Van der Waals corrections

vdw_corr	CHARACTER
<i>Default:</i>	'none'
<i>See:</i>	london_s6 , london_rcut , london_c6 , london_rvdw , dftd3_version , dftd3_threebody , ts_vdw_econv_thr , ts_vdw_isolated , xdm_a1 , xdm_a2
Type of Van der Waals correction. Allowed values:	
'grimme-d2', 'Grimme-D2', 'DFT-D', 'dft-d' : Semiempirical Grimme's DFT-D2. Optional variables: london_s6 , london_rcut , london_c6 , london_rvdw S. Grimme, J. Comp. Chem. 27, 1787 (2006), doi:10.1002/jcc.20495 V. Barone et al., J. Comp. Chem. 30, 934 (2009), doi:10.1002/jcc.21112	
'grimme-d3', 'Grimme-D3', 'DFT-D3', 'dft-d3' : Semiempirical Grimme's DFT-D3. Optional variables: dftd3_version , dftd3_threebody S. Grimme et al, J. Chem. Phys 132, 154104 (2010), doi:10.1063/1.3382344	
'TS', 'ts', 'ts-vdw', 'ts-vdW', 'tkatchenko-scheffler' : Tkatchenko-Scheffler dispersion corrections with first-principle derived C6 coefficients. Optional variables: ts_vdw_econv_thr , ts_vdw_isolated See A. Tkatchenko and M. Scheffler, PRL 102, 073005 (2009) .	
'MBD', 'mbd', 'many-body-dispersion', 'mbd_vdw' : Many-body dispersion (MBD) correction to long-range interactions. Optional variables: ts_vdw_isolated A. Ambrosetti, A. M. Reilly, R. A. DiStasio, A. Tkatchenko, J. Chem. Phys. 140 18A508 (2014).	
'XDM', 'xdm' : Exchange-hole dipole-moment model. Optional variables: xdm_a1 , xdm_a2 A. D. Becke et al., J. Chem. Phys. 127, 154108 (2007), doi:10.1063/1.2795701 A. Otero de la Roza et al., J. Chem. Phys. 136, 174109 (2012), doi:10.1063/1.4705760	
Note that non-local functionals (eg vdw-DF) are NOT specified here but in input_dft	

Check the online documentation at https://www.quantum-espresso.org/Doc/INPUT_PW.html

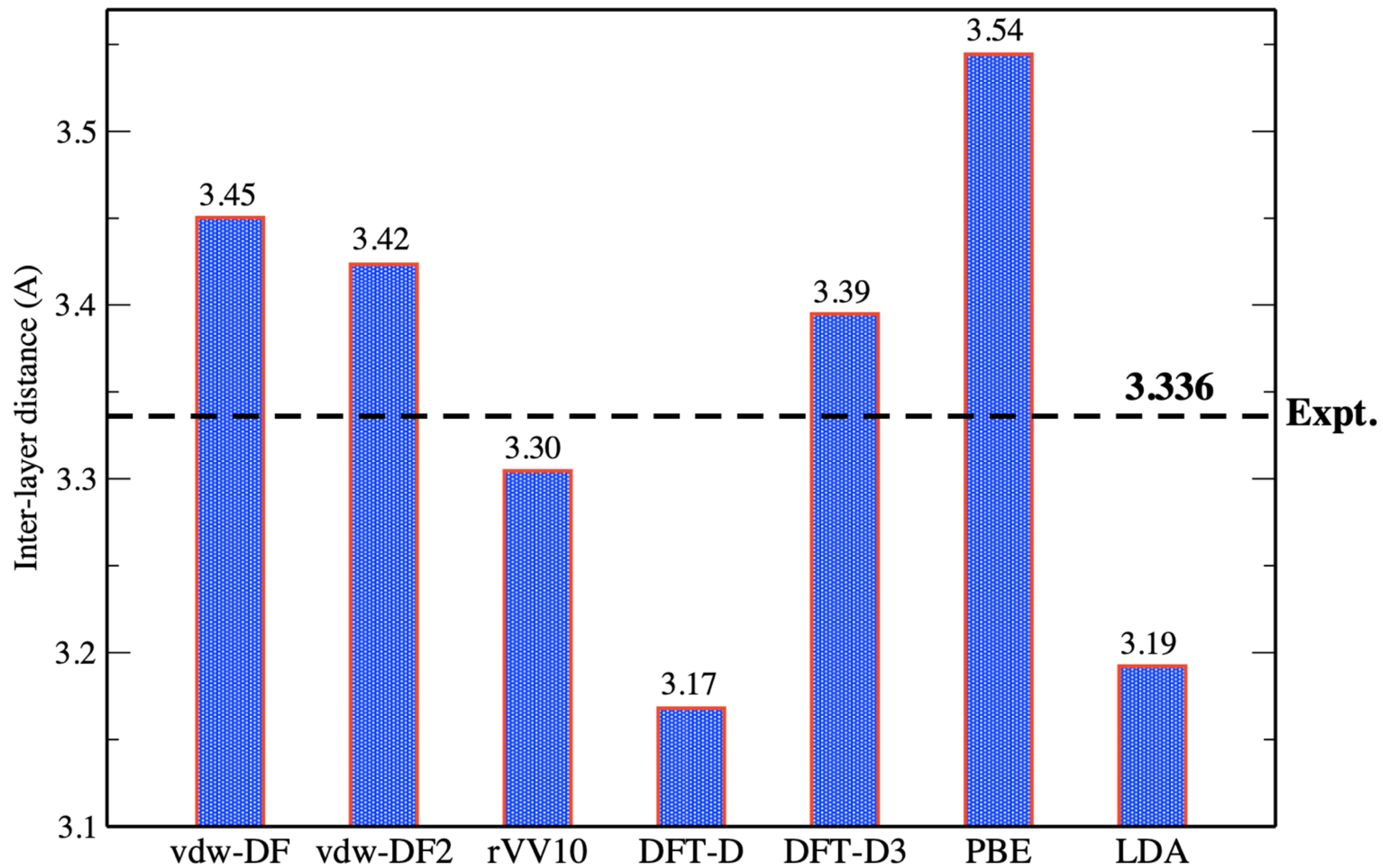
Structural optimization of graphite

Study different cases:

1. input_dft = 'vdw-DF'	@PBE	pseudo	(non-local)	C.pbe-rrkjus.UPF
2. input_dft = 'vdw-DF2'	@PBE	pseudo	(non-local)	C.pbe-rrkjus.UPF
3. input_dft = 'rVV10'	@PBE	pseudo	(non-local)	C.pbe-rrkjus.UPF
4. vdw_corr = 'DFT-D'	@PBE	pseudo	(semi-empirical)	C.pbe-rrkjus.UPF
5. vdw_corr = 'DFT-D3'	@PBE	pseudo	(semi-empirical)	C.pbe-rrkjus.UPF
6. Normal PBE calculation	@PBE	pseudo		C.pbe-rrkjus.UPF
7. Normal LDA calculation	@LDA	pseudo		C.pz-rrkjus.UPF

Use XCrySDen (outside the Docker) to determine the inter-layer distances after the optimization

Inter-layer distance in graphite



Note: these results must be carefully converged (cutoff, k points, etc.)

Exercise 3

meta-GGA for Si and Fe

Input file

```
&control
  calculation='scf'
  restart_mode='from_scratch',
  prefix='Si'
  pseudo_dir = '../..../pseudo'
  outdir='./tmp'
  verbosity='high'
/
&system
  ibrav = 2,
  celldm(1) = 10.262,
  nat = 2,
  ntyp = 1,
  ecutwfc = 40.0,
  input_dft = 'SCAN'
  nbnd = 5
/
&electrons
  conv_thr = 1.d-10
/
ATOMIC_SPECIES
Si 28.086 Si.SCAN.UPF
ATOMIC_POSITIONS {alat}
Si 0.00 0.00 0.00
Si 0.25 0.25 0.25
K_POINTS {automatic}
12 12 12 0 0 0
```

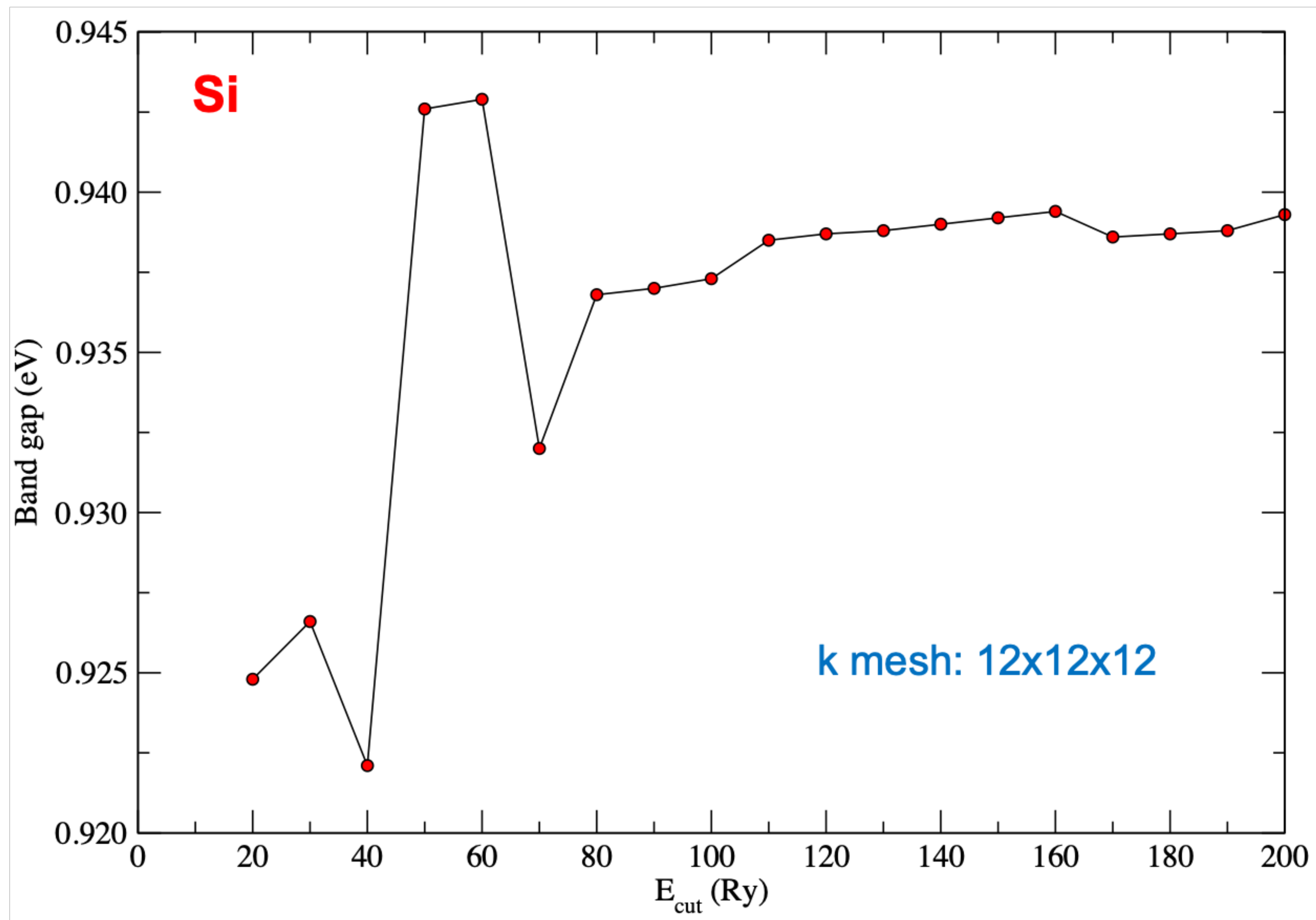
Experimental lattice parameter

It is necessary to perform convergence tests w.r.t ecutwfc

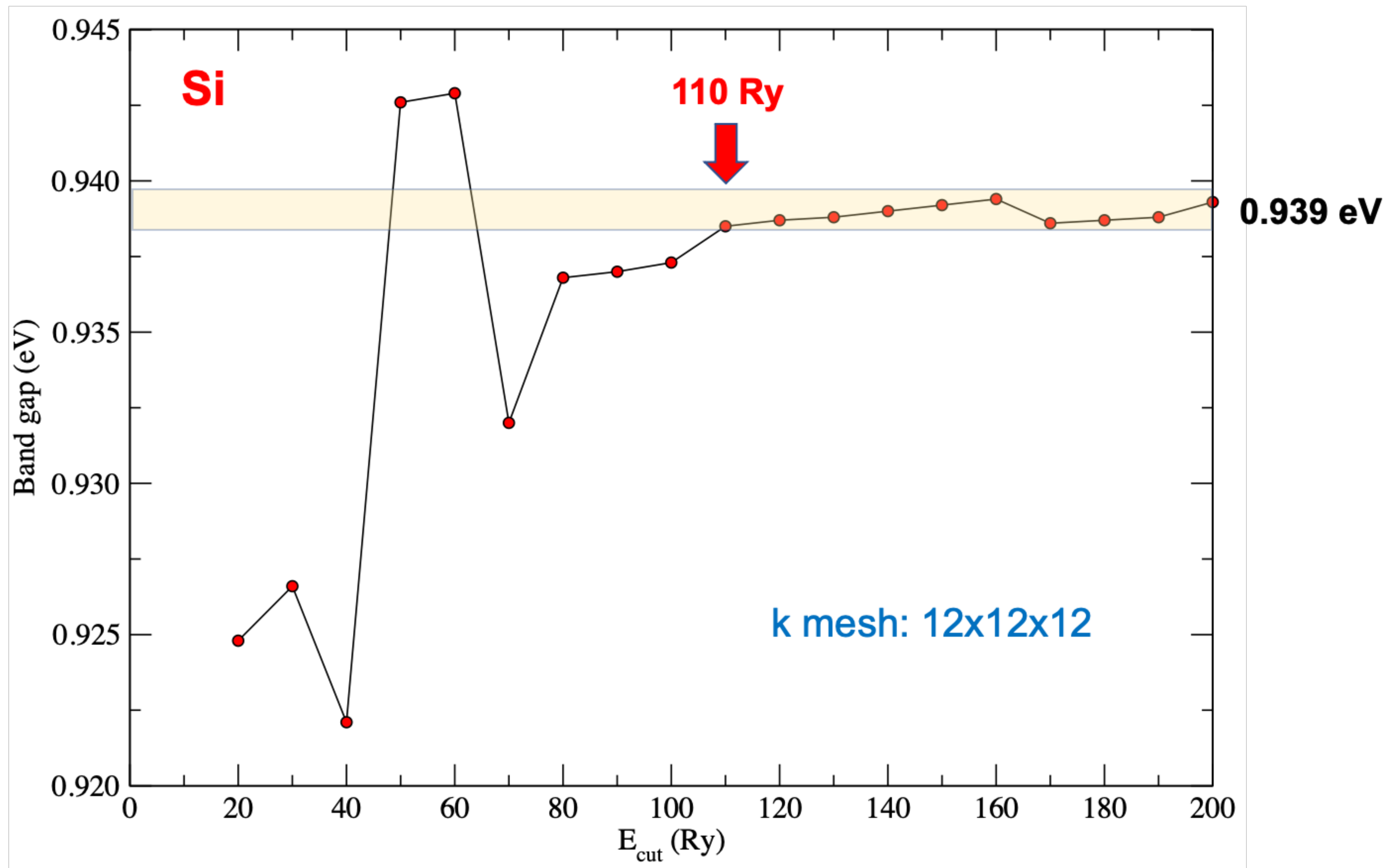
Set up the SCAN functional from the input
Quantum ESPRESSO must be compiled with Libxc

Pseudopotential generated using the SCAN functional
<https://yaoyi92.github.io/scan-tm-pseudopotentials.html>

SCAN functional & SCAN pseudopotential



SCAN functional & SCAN pseudopotential



SCAN functional & PBE pseudopotential

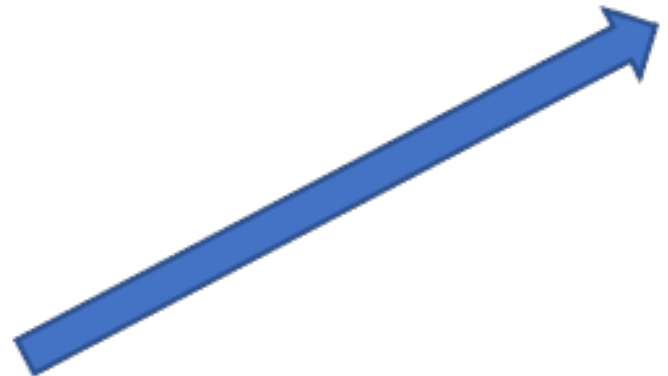
```
&control
  calculation='scf'
  restart_mode='from_scratch',
  prefix='Si'
  pseudo_dir = '../..../pseudo'
  outdir='./tmp'
  verbosity='high'
/
&system
 ibrav = 2,
  celldm(1) = 10.262,
  nat = 2,
  ntyp = 1,
  ecutwfc = 110.0,
  input_dft = 'SCAN'
  nbnd = 5
/
&electrons
  conv_thr = 1.d-10
/
ATOMIC_SPECIES
Si 28.086 Si.pbe_PseudoDojo.UPF
ATOMIC_POSITIONS {alat}
Si 0.00 0.00 0.00
Si 0.25 0.25 0.25
K_POINTS {automatic}
12 12 12 0 0 0
```

Pseudo Dojo library
<http://www.pseudo-dojo.org>



```
&control
  calculation='scf'
  restart_mode='from_scratch',
  prefix='Si'
  pseudo_dir = '../..../pseudo'
  outdir='./tmp'
  verbosity='high'
/
&system
 ibrav = 2,
  celldm(1) = 10.262,
  nat = 2,
  ntyp = 1,
  ecutwfc = 110.0,
  input_dft = 'SCAN'
  nbnd = 5
/
&electrons
  conv_thr = 1.d-10
/
ATOMIC_SPECIES
Si 28.086 Si_ONCV_PBE-1.2.upf
ATOMIC_POSITIONS {alat}
Si 0.00 0.00 0.00
Si 0.25 0.25 0.25
K_POINTS {automatic}
12 12 12 0 0 0
```

SG15 library



http://www.quantum-simulation.org/potentials/sg15_oncv/

SCAN functional with different pseudopotentials

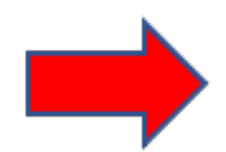
Ecut = 110 (Ry), k mesh: 12x12x12

Functional	Pseudopotential	Library	Gap (eV)
SCAN	SCAN	Yi Yao's library	0.94
SCAN	PBE	SG15 ONCV	0.83
SCAN	PBE	Pseudo Dojo	0.64

Gap_{expt} = 1.17 (eV)

SCAN functional with different pseudopotentials

Ecut = 110 (Ry), k mesh: 12x12x12



Functional	Pseudopotential	Library	Gap (eV)	
SCAN	SCAN	Yi Yao's library	0.94	NLCC=.false.
SCAN	PBE	SG15 ONCV	0.83	NLCC=.false.
SCAN	PBE	Pseudo Dojo	0.64	NLCC=.true.

Gap_{expt} = 1.17 (eV)

Currently, nonlinear core correction (NLCC) is not implemented for meta-GGA in Quantum ESPRESSO!

Pseudopotentials that have NLCC=.true. introduce some inconsistency for meta-GGA calculations.

SCAN functional with different pseudopotentials

Ecut = 110 (Ry), k mesh: 12x12x12

Functional	Pseudopotential	Library	Gap (eV)
SCAN	SCAN	Yi Yao's library	0.94
SCAN	PBE	SG15 ONCV	0.83
SCAN	PBE	Pseudo Dojo	0.64
PBE	PBE	SG15 ONCV	0.56
PBE	PBE	Pseudo Dojo	0.57

Gap_{expt} = 1.17 (eV)

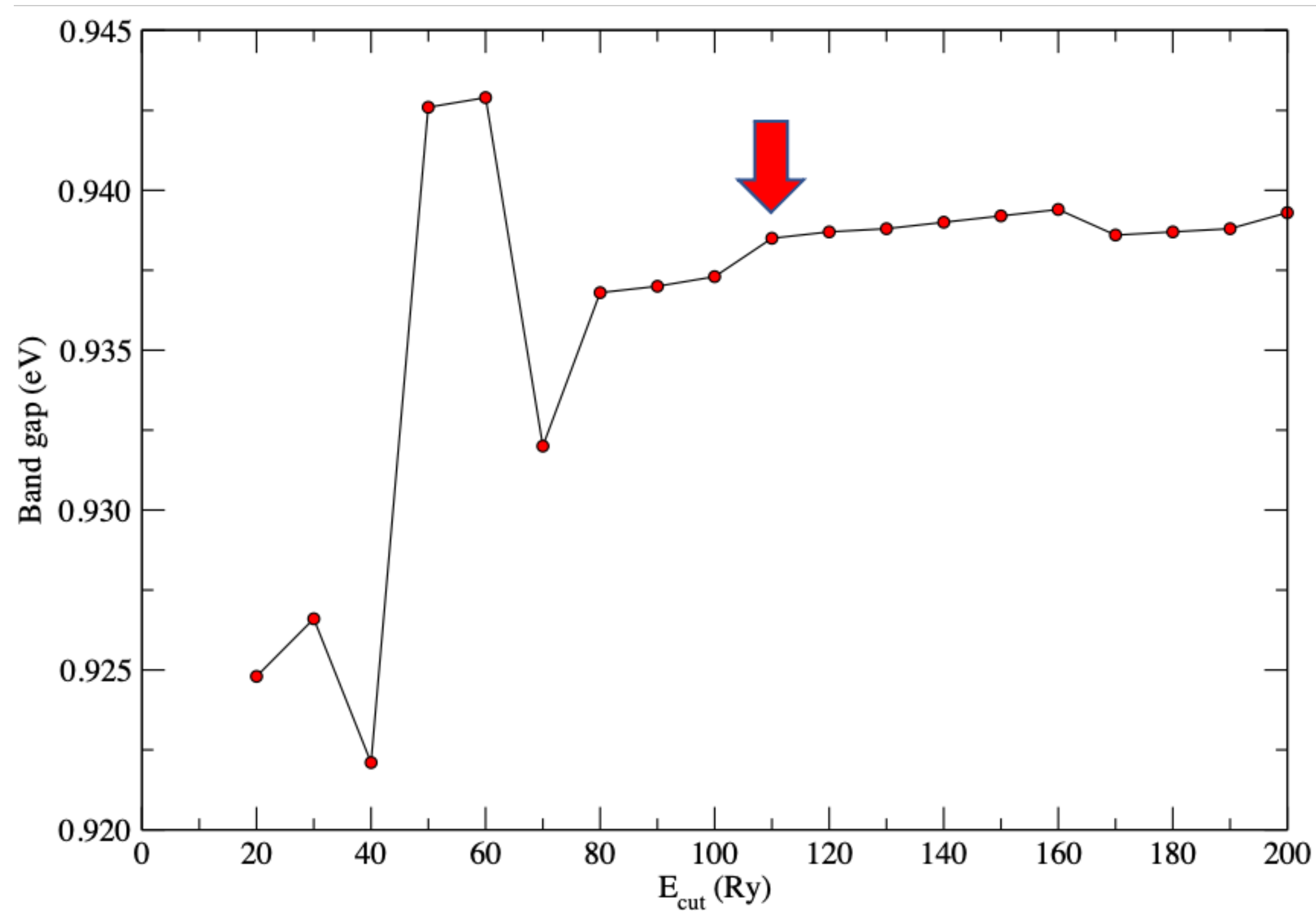
SCAN calculation with the SCAN pseudopotential gives the most accurate results

Y. Yao and Y. Kanai, J. Chem. Phys. 146, 224105 (2017).

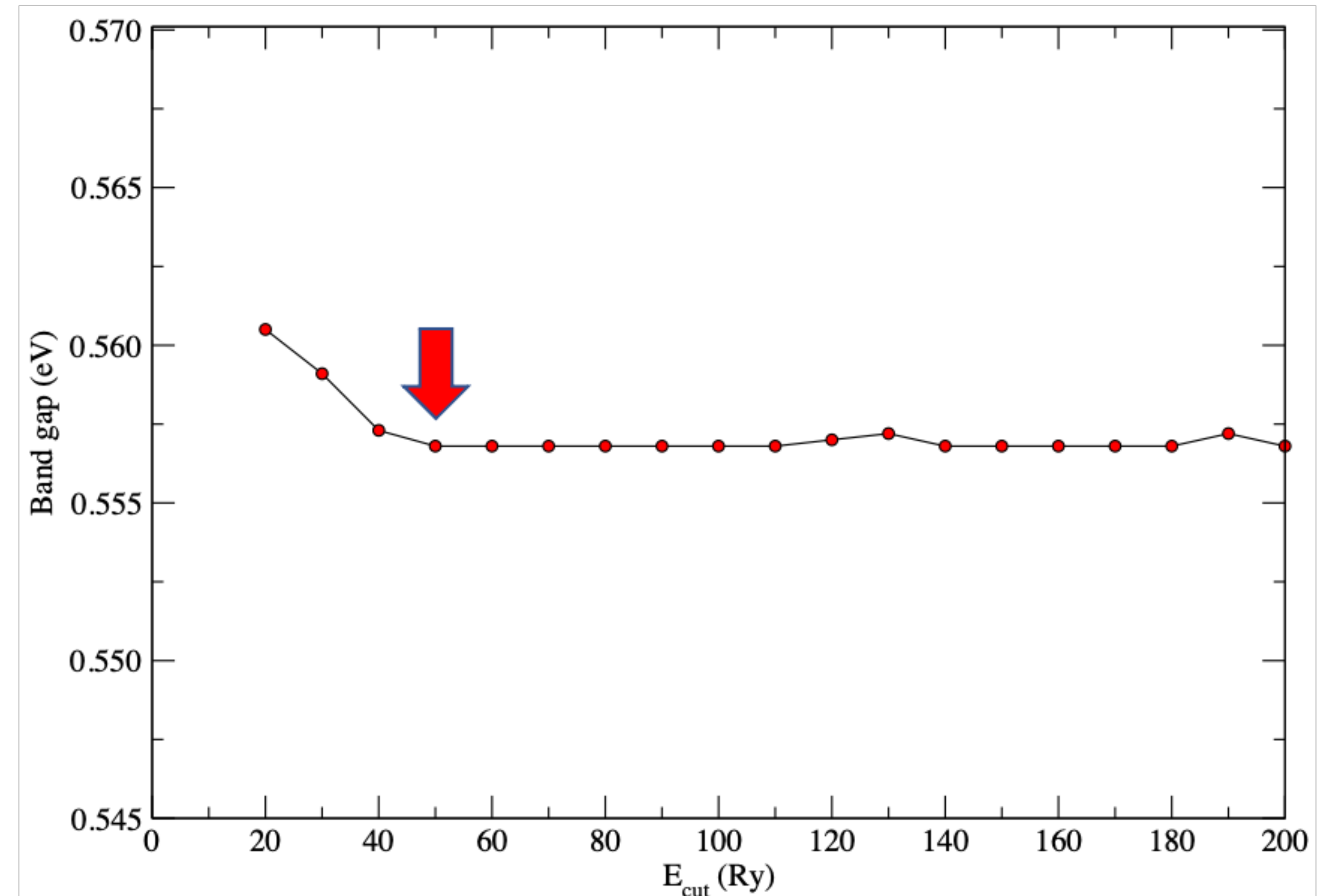
Convergence with respect to E_{cut}

k mesh: 12x12x12

SCAN with SCAN pseudopotential

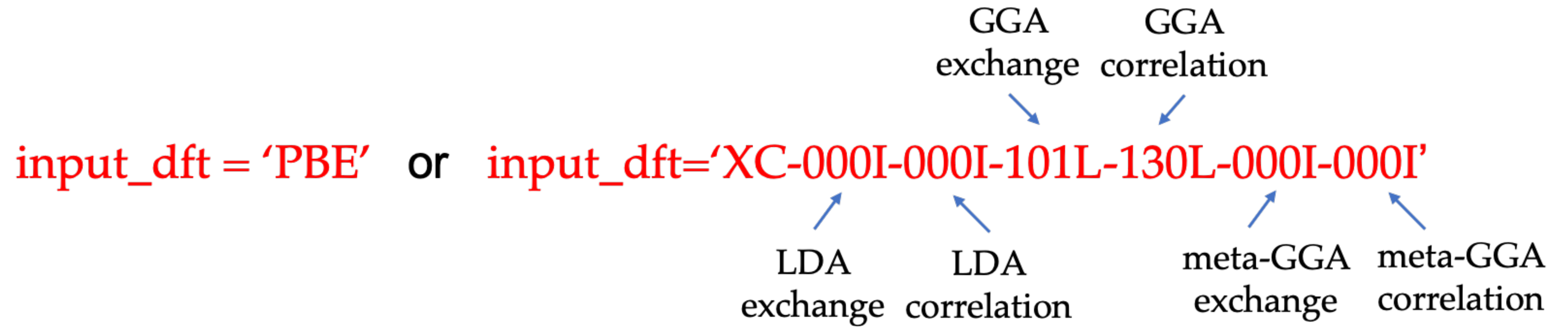


PBE with PBE pseudopotential (SG15)



Convergence is achieved much faster with PBE than with SCAN

Setting up the XC functional



Setting up the XC functional

input_dft = 'PBE' or input_dft='XC-000I-000I-101L-130L-000I-000I'

The diagram illustrates the mapping of various density functional theory (DFT) functionals to a specific input string. The string is 'XC-000I-000I-101L-130L-000I-000I'. Above the string, 'GGA exchange' points to the first '000I' and 'GGA correlation' points to the second '000I'. Below the string, 'LDA exchange' points to the first '000I', 'LDA correlation' points to the second '000I', 'meta-GGA exchange' points to the '101L', and 'meta-GGA correlation' points to the '130L'.

```
input_dft = 'SCAN' or input_dft='XC-000I-000I-000I-000I-263L-267L'
```

~~input_dft = 'rSCAN'~~ or input_dft='XC-000I-000I-000I-000I-493L-494L'

`input_dft = 'r2SCAN'` or `input_dft='XC-000I-000I-000I-000I-497L-498L'`

no short
name
in QE

Read more about this here: https://www.quantum-espresso.org/Doc/user_guide/node13.html

Libxc with IDs: <https://libxc.gitlab.io/>

Different flavors of SCAN

Functional	Pseudopotential	Library	Gap (eV)
SCAN	PBE	SG15 ONCV	0.83
rSCAN	PBE	SG15 ONCV	0.71
r ² SCAN	PBE	SG15 ONCV	0.70

$$\text{Gap}_{\text{expt}} = 1.17 \text{ (eV)}$$

SCAN is more accurate than rSCAN and r²SCAN for predicting the band gap of bulk Si
(at least when using the PBE pseudopotential)

Input file

```
&control
  calculation='scf'
  restart_mode='from_scratch',
  prefix='Fe'
  pseudo_dir = '../..//pseudo'
  outdir='./tmp'
  verbosity='high'
/
&system
 ibrav = 3,
  celldm(1) = 5.418,
  nat = 1,
  ntyp = 1,
  ecutwfc = 50.0,
  occupations = 'smearing',
  smearing = 'mv',
  degauss = 0.02,
  nspin = 2,
  starting_magnetization(1) = 0.5,
  input_dft = 'SCAN'
/
&electrons
  conv_thr = 1.d-10
/
ATOMIC_SPECIES
Fe 55.845 Fe_ONCV_PBE-1.2.upf
ATOMIC_POSITIONS {crystal}
Fe 0.000000 0.000000 0.000000
K_POINTS {automatic}
20 20 20 0 0 0
```

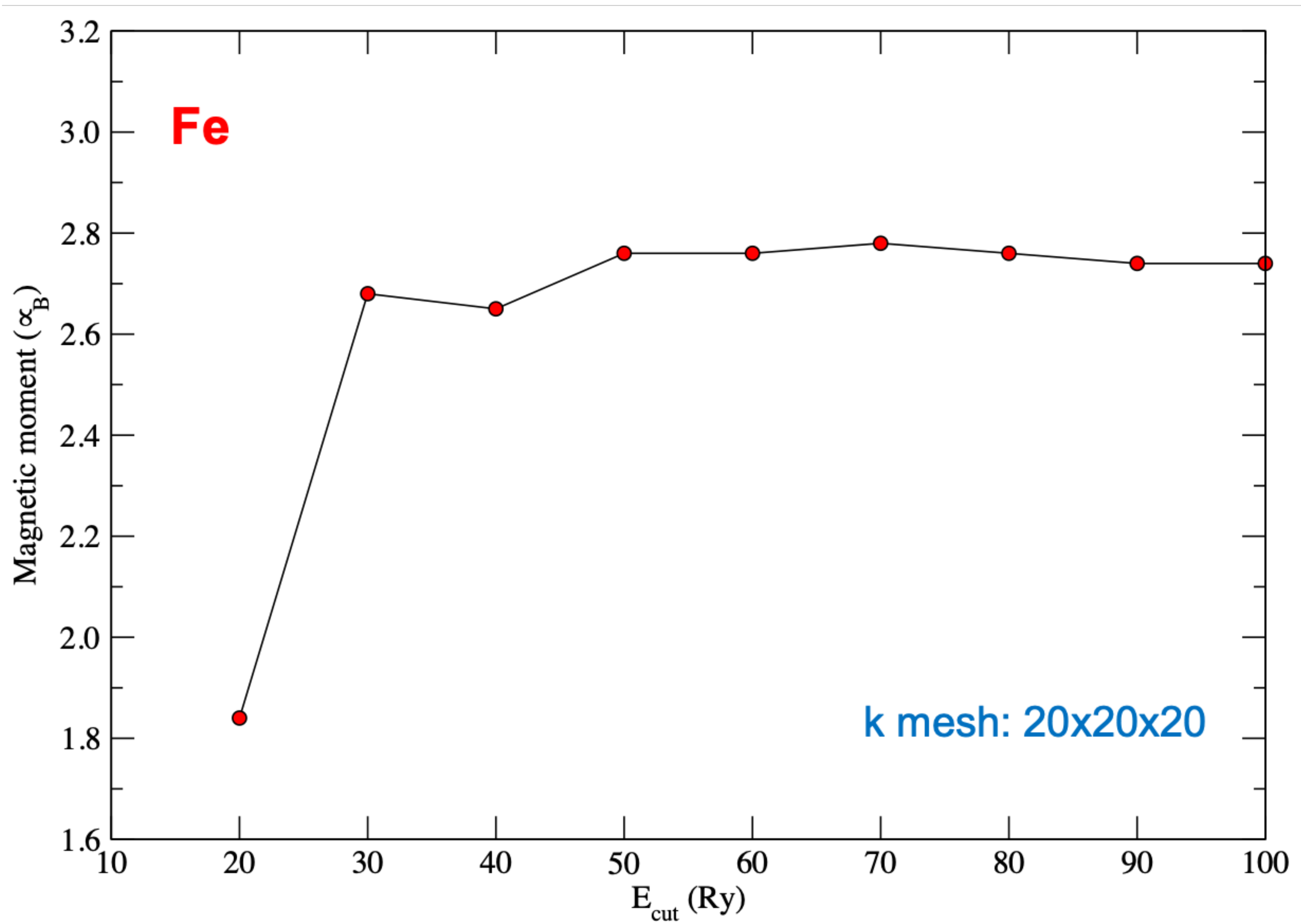
Experimental lattice parameter

It is necessary to perform convergence tests w.r.t ecutwfc

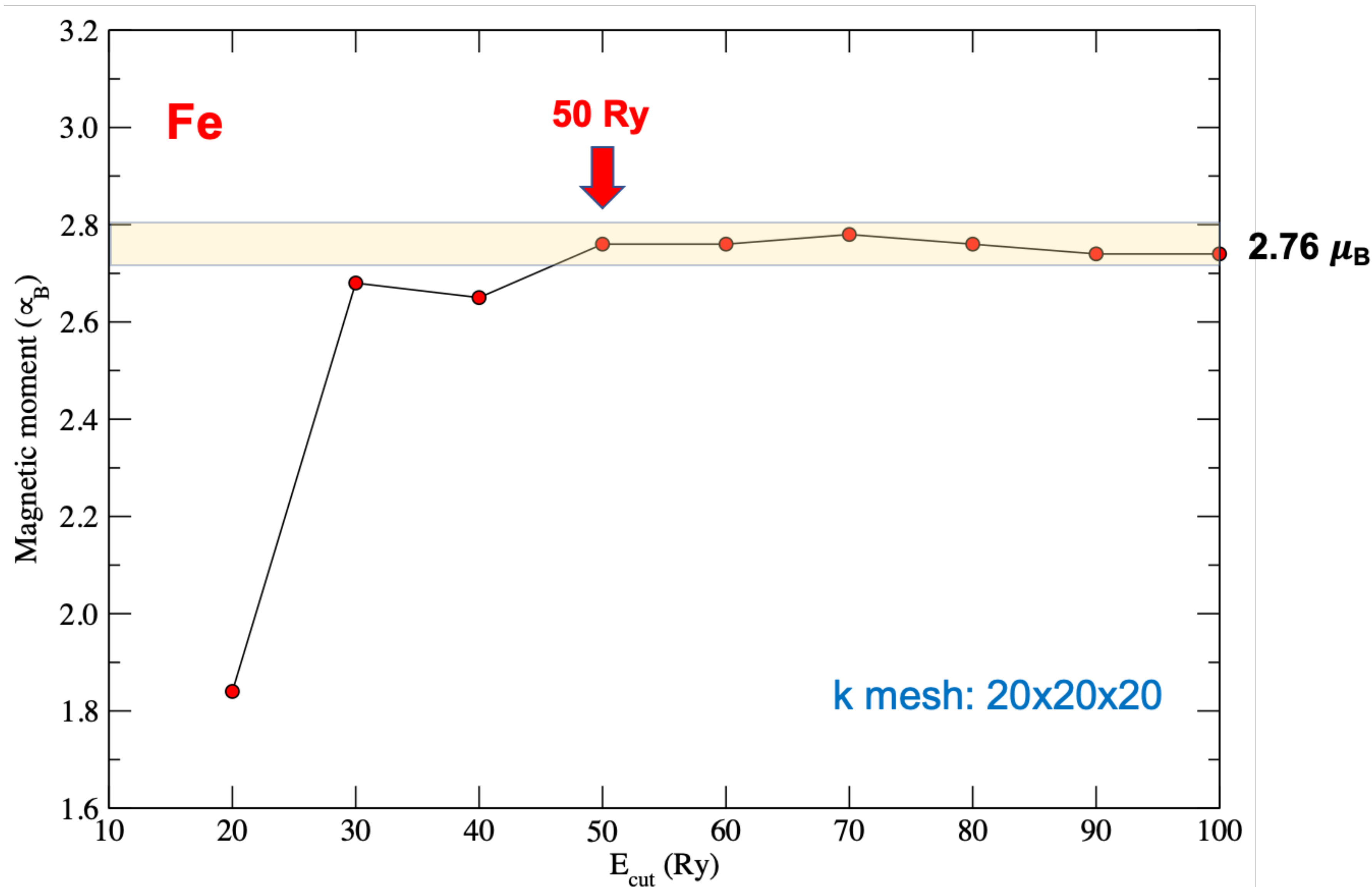
Set up the SCAN functional from the input
Quantum ESPRESSO must be compiled with Libxc

Pseudopotential generated using the PBE functional
SG15 ONCV library (there is no SCAN pseudo for Fe)

SCAN functional & PBE pseudopotential (SG15)

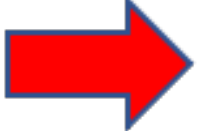


SCAN functional & PBE pseudopotential (SG15)



SCAN functional with different pseudopotentials

Ecut = 50 (Ry), k mesh: 20x20x20

WARNING 

Functional	Pseudopotential	Library	$m (\mu_B)$
SCAN	PBE	SG15 ONCV	2.76
SCAN	PBE	Pseudo Dojo	2.67
PBE	PBE	SG15 ONCV	2.28
PBE	PBE	Pseudo Dojo	2.26

NLCC=.false.

NLCC=.true.

$$m_{\text{expt}} = 1.98 - 2.13 (\mu_B)$$

SCAN overestimates magnetic moments in itinerant ferromagnets

PBE is in closer agreement with experiments than SCAN for magnetic moments of itinerant ferromagnets

M. Ekholm et al., Phys. Rev. B 98, 094413 (2018).

Different flavors of SCAN

Functional	Pseudopotential	Library	$m (\mu_B)$
SCAN	PBE	SG15 ONCV	2.76
rSCAN	PBE	SG15 ONCV	2.73
r ² SCAN	PBE	SG15 ONCV	2.73

$$m_{\text{expt}} = 1.98 - 2.13 (\mu_B)$$

Different flavors of SCAN give very similar magnetic moments for bulk Fe